

Signal Acquisition and Analysis Using MATLAB



PROF. K. PADMANABHAN AND
DR S. ANANTHI

You can capture an input signal, observe its waveform and variation with time on an oscilloscope, but further processing of the signal is not feasible with it. When we need to digitise signals into samples and observe the waveform through plotting, we need a computer along with software. The software such as LabVIEW can be used to acquire signals through an analogue-to-digital converter (ADC) module and analyse them. For example, by capturing a musical sequence of notes, one can study the spectrum of the signal to get more information.

Presented here is a very simple method to acquire audio signals from a transducer (such as a microphone) or circuit under test and display them on a computer screen through MATLAB software for further analysis. There is no need for any ADC chip or module even, but the signals should be within five volts peak and within 40 to 20,000Hz in frequency content.

The hardware required for this purpose is nothing more than an audio connector, which is inserted into the Line input socket of the sound card available on a computer. Present computer motherboards have direct audio interface connectors on them, which are available on the back panel of the desktops (Fig. 1). Similarly in laptops, audio connectors for input and output are available, either on the side or front edges.

Software program

The computer must be installed with MATLAB version 6.5 or later. The enormous power of MATLAB software, which has excellent graphical display capabilities, processing commands for Fourier transforms, filtering, statistical parameters, etc., is available to the user

for doing anything he likes with the signal he interfaces through the sound card socket.

Two analogue signals can be interfaced to the PC. The stereo Line-in input pin of the sound socket is

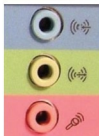


Fig. 1: MIC-In and Line-In inputs normally available on the PC

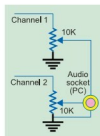


Fig. 2: Two-channel signal inputs through potentiometers

connected to the signals through two 10-kilo-ohm potentiometers (Fig. 2) to adjust amplitude levels, as and when necessary. A shielded-wire and ganged-linear carbon film potentiometer may be used along with an audio jack.

Interfacing sound card and MATLAB. The audio input always has a pair of signals for use in stereo operation. Thus, two signals can be simultaneously captured by the sound card and then analysed with the MATLAB program.

One of the signals can act as a starting trigger input, after which the second channel signal is acquired. The duration of acquiring a signal must be specified in the program. It can last for up to several seconds—even minutes in some cases.

All audio interface devices in a PC or laptop use a sampling rate for the analogue inputs at 44,100Hz, which limits the upper frequency of signal to 20kHz. Most signals we deal with are within this range.

After making required connections to the PC, run the MATLAB program from Windows XP (it may not run in a 64-bit machine). The program acquires analogue signals for a specified duration and makes the data samples available instantly. The waveform of the signals can be displayed by the plot command.

To see the spectrum of the signal, the FFT command is used and the display of the spectrum can be seen by plotting its absolute value. The MATLAB code for the same is listed below:

```
AI=analoginput('winsound',0)
addchannel(AI,[1 2])
```

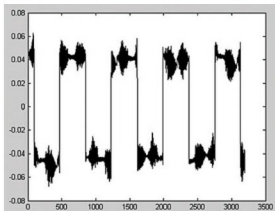


Fig. 3: Acquired data from a square wave input from a 555 circuit at about 60Hz

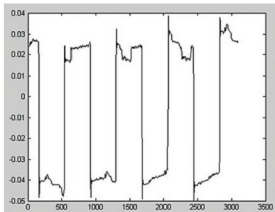


Fig. 4: The filtered graph of the acquired data shown in Fig. 3